

Faithful Baselines, Six Dead Mechanisms, and One Diagnosis: A Measurement Study of Streaming Video-LLM Memory

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Abstract

Streaming video-LLM memory is a crowded design space — multi-grained compression, KV retrieval, query-agnostic pruning, trained streaming architectures — evaluated with incomparable judges, unmatched configurations, and no shared baselines. We built a controlled evaluation stack (faithful reproductions of MuKV and ReKV, a locally hosted 72B judge with a measured systematic offset that is stable across three independent systems, pre-registered gates, and multiplicity correction applied symmetrically to our own claims) and used it to test six candidate mechanisms drawn from the 2025–2026 literature. All six died under pre-registration; each death is a measurement with a root cause. Two methodological contributions survive: an oracle-null protocol that shows a widely-cited style of clairvoyant upper bound is a max-over-policies artifact, and a ceiling decomposition of every all-wrong question across three benchmarks attributes failures to model capability at long temporal windows rather than retrieval or storage — with a powered causal test then refuting the one plausible lever (window coverage), closing the mechanism space on evidence rather than on insufficient power. The field, in our measurement, is optimizing second-order terms while the first-order term goes unmeasured. We provide the full reproducibility trail (every number traces to a judged artifact, config, and code state) and the Pareto accounting of where the accuracy–storage frontier actually stands. Our final claim is deliberately modest: at true 0.5 fps, a multi-grain position-disentangled store matches CVPR’26 SOTA accuracy at comparable storage with a different architecture, and the interesting problems are elsewhere.

1 Introduction

Streaming video understanding — a model watches an unbounded stream and answers questions about it later — has produced a dense mechanism literature in two years: KV-cache retrieval [3], multi-grained compression [11], query-agnostic pruning [1, 6, 14], hierarchical and trained memory [12, 17, 18], token reduction [2, 4, 5, 10, 19], and agentic memory [13, 15]. Every paper reports wins. Almost none of them can be compared: judges differ (GPT-3.5 vs GPT-4 class vs proprietary APIs), anchor configurations differ (a CVPR’26 system’s published RVS number is produced by a config that is not its repository’s default), effective sampling rates are rarely asserted, and clairvoyant “oracle” bounds are quoted without null models.

This paper is a measurement study of that landscape, conducted while attempting to beat its strongest representative. We built a controlled stack (Section 2): faithful on-hardware reproductions of MuKV and ReKV, a locally hosted 72B judge whose systematic offset (≈ -9 accuracy points vs

published GPT-judged numbers) is stable across three independent systems, a standing effective-fps assertion, and a pre-registration discipline in which every gate specifies its pass criteria, its minimum detectable effect, and its multiplicity family before numbers exist.

Against this stack we tested *six mechanisms* drawn from the literature’s own admitted openings (Section 3): differential (residual) KV coding, content-adaptive GOP sizing, retrieval-adaptive precision, write-time debiasing, deferred query-time commitment, and a learned commitment policy. **All six failed under pre-registration**, and each failure carries a root cause: adjacent-frame KV residuals encode rotary position drift, not content change; quantization sensitivity is flat across retrieval frequency; entropy-triggered refinement is inert; per-question policy headroom is a max-over-policies artifact (Section 4); and the coverage experiment designed to confirm our own diagnosis returned underpowered-null at $n=83$.

Two things survive the carnage. First, a methodological contribution: the *oracle-null protocol* (Section 4), which shows that clairvoyant per-question selection bounds — used across the literature to motivate retrieval and allocation mechanisms — are largely artifacts of maximizing over exchangeable policies, and provides the permutation machinery to say so. Second, a diagnosis (Section 5): a ceiling decomposition of every all-wrong question on three benchmarks finds *zero* retrieval failures and a dominant model-capability failure mode concentrated in long-window temporal grounding; the coverage–planner-gain correlation that initially looked causal was *refuted on evidence* by a powered pooled regression — coverage is not the lever, and the field, in our measurement, is optimizing second-order terms (which blocks to select) while the first-order term (whether the model can ground a long temporal window at all) goes unmeasured.

Section 6 gives the honest frontier: after correcting our own frame-rate bug, DECAF’s multi-grain store matches MuKV’s faithful accuracy (paired $p=0.203$) and full-fidelity ReKV ($p=0.139$) at $8.4\text{--}11.4\times$ MuKV’s published storage in fp16, with the int4-group-scale repair measured separately. We claim no accuracy win — none survives correction — and we show where one might actually live.

Contributions.

- (1) A controlled evaluation stack for streaming video-LLM memory: faithful MuKV/ReKV reproductions, a stability-characterized local judge, fps assertions, and a pre-registration template (gates, MDE, multiplicity family) reusable beyond this study.
- (2) Six mechanism evaluations with root-cause post-mortems, each reproducible from its run directory.

- (3) The oracle-null protocol: a permutation framework establishing that clairvoyant selection bounds are max-artifacts, with three-benchmark evidence.
- (4) A coverage-planner-gain correlation and ceiling decomposition identifying long-window temporal grounding as the binding constraint.
- (5) A corrected Pareto accounting and an explicit map of the design space's dead ends.

2 The Controlled Evaluation Stack

2.1 Faithful reproductions

We reproduce two reference systems on one 8×RTX 6000 Ada server (4 GPUs available, CUDA 12.6): ReKV [3] (commit 1fd9a3d + Volta-port patches) and MuKV [11] (2127b875). Gate 0 of our study was MuKV's own reproduction: its repository's argparse defaults are *not* its paper's RVS configuration — the published number (57.9 Ego, 7B) requires the segment-dominant keep-ratio setting found only in its shell script. Running the defaults yields 46.0 in our harness; the paper configuration yields 50.9, within 0.5 points of the offset-adjusted expectation (Appendix A). Code path was independently validated by a minimal-pruning run reaching ReKV-level accuracy. **Lesson institutionalized: runner defaults are not paper configs; every arm encodes its configuration explicitly.**

2.2 The judge and its stability

OpenAI APIs are unreachable from our infrastructure; we judge with a locally hosted Qwen2.5-72B (vLLM, TP=4, temperature 0, the RVS grading prompt verbatim from the evaluation codebase). The judge carries a systematic offset versus published GPT-judged numbers — and the offset is the contribution: measured across three independent reproductions it is −9.2 (ReKV-0.5B), −9.6 (ReKV-7B), and −8.6 (MuKV-7B). A stable offset across independent systems is far stronger evidence for *relative* comparability than any single reproduction: within-harness orderings and margins are trustworthy even where absolute numbers are not. Two local judges (72B, 14B) agree at 94% on a fixed prefix. All letter-based cross-benchmark evaluations use deterministic exact-match scoring, eliminating the judge entirely.

2.3 Measurement discipline

Three standing rules, each born from a documented failure: (i) *effective-fps assertion* — every video arm logs and asserts its effective frame rate from frame counts (a double-resampling bug had silently halved our rate against 0.5-fps anchors; Section 6); (ii) *label-at-emit* — every statistical output carries its test, tail, and complement in the label (a mislabeled tail-complement cost two correction rounds); (iii) *pre-registration* — every experiment declares its primary comparison, multiplicity family, and minimum detectable effect before numbers exist. Multiplicity correction (Holm; BH reported) is applied *symmetrically*: it killed our own headline claim (Section 6) exactly as it would a reviewer's.

2.4 Benchmarks

RVS-Ego (10 Ego4D streams, 1,465 questions, GPT-judged open-ended) [3, 16]; StreamingBench real-time split and OVO-Bench real-time mode (both deterministic letter-match; subsets of 180 questions matched in statistical weight, long-horizon videos preferred) [7, 8]. Ground truth for coverage and temporal analysis comes from each benchmark's annotations.

3 Six Mechanisms, Six Post-Mortems

Each mechanism below was proposed from the literature's own admitted openings, built, trap-verified, and killed by a pre-registered gate. Full run directories (config, environment, code state, metrics, judged artifacts) accompany each.

3.1 M1–M3: the inside-the-model trio (all dead)

Differential KV coding (residuals against keyframes, codec-style) died at its gate: pre-RoPE key residuals are 0.25–0.46 of block energy (values 0.54–0.94), because stored KV is a function of the whole prefix — adjacent-frame difference measures rotary position drift, not content change (differencing post-RoPE keys would inflate residuals 73×). Compression at matched reconstruction error reached 1.62 GB/h — 0.56× of MuKV's published rate, where the gate needed 3×, and ~1.0× over trivially quantizing every frame independently. **Adaptive GOP sizing** died with it: threshold-triggered GOP length ratios across static/dynamic regimes were 0.74–1.11 at every threshold — fixed-stride keyframing in disguise, with no event-boundary alignment. **Retrieval-adaptive precision** died on two measurements: 2-bit quantization distortion is *flat* across selection-frequency quintiles (Hot/Cold ratio 0.99, gate 1.5), and 4-bit is near-lossless at logit level everywhere — with no sensitivity gradient, adaptive bit-width has nothing to allocate (see TB-findings in Appendix B for the answer-level int4 caveat).

3.2 M4–M5: position disentanglement and write-time debiasing

Position-disentangled store (pre-RoPE content keys + position sidecar) was built and survived trap-verification, with one correction: exact-position fetch at true streaming positions (~86k tokens) collapses generation — answers truncate mid-sentence late-stream, the signature of rotary out-of-distribution. Rank-remapped fetch restores bit-exact stock behavior (10/10 character-identical at the failure scale; benchmark delta 0.0). The literature's re-indexing patchwork [9, 14, 17] reads, in hindsight, as partial fixes to this same fetch-position problem. **Write-time debiasing** measurably works at the statistic level (attention-vs-position residual 0.412 → 0.330) and measurably does nothing at the answer level (paired $p=0.077$): MuKV's positional bias is real, and removing it does not move RVS-Ego accuracy.

3.3 M6: deferred query-time commitment (the ambitious one)

The flagship idea — store a hedged superset at write time, commit grains per query with cross-grain consistency and entropy-triggered refinement — failed at every valid configuration: deferred beat write-time commitment by +0.8 (tax 0.5) and +1.6 (tax 0.75, later shown degenerate-by-quantization), both paired- $p \approx 0.3$ under a pre-declared gate of +2. The entropy second pass is inert (fires 10.8%, zero gain where it fires). The hedged superset has no operating point below tax 0.5 (a per-signal frame-quantization dead zone, kept visible as a known limitation). The system’s single-grain ablations do validate one design choice: no single grain suffices (patch-only 45.6 / frame-only 43.4 / segment-only 41.1 vs the multi-grain system ~ 52).

3.4 The seventh candidate that never ran

A learned commitment scorer was the planned successor to M6. A pre-registered learnability probe — can a trivial selector predict the winning grain policy from question metadata and store statistics? — returned ~ 0 held-out gain (200 seeded resplits: mean -1.02 ± 1.75 pt; 26% of splits positive), and the oracle analysis (Section 4) shows the headroom it would harvest is an artifact. The mechanism was abandoned *before* implementation, on evidence — the cheapest possible kill.

4 The Oracle-Null Protocol

4.1 The artifact we set out to harvest

Deferred commitment was motivated by a clairvoyant bound: per-question best-grain selection scored 51.6 on the RVS subset vs the system’s 41–43 — an apparent +9 accuracy ceiling. Because decoding is greedy with fixed seed, that number is deterministic and looks attainable. A learnability probe failed to capture it from cheap features; this section explains why the number itself misleads.

4.2 The null model

Under a within-question exchangeability null — permuting the four policies’ per-question correctness labels — a *random per-question assignment* already achieves oracle-minus-best-fixed gaps of 7.8 ± 0.6 (RVS) and 5.6 ± 0.7 (StreamingBench). The observed gaps (+6.3/+5.0/+4.4 across three benchmarks) sit at the 1.1th, 12.3th, and 2.5th percentiles of their nulls: **the clairvoyant margin is fully explained by max-over-exchangeable-policies selection**. Reporting it as attainable headroom — a widespread practice — is a category error the null makes precise.

4.3 Implications

First, every mechanism in Section 3 that targeted retrieval/allocation “headroom” was chasing an artifact; the oracle-null protocol would have shown this in hours. Second, the protocol is cheap and general: it requires only per-question correctness vectors from a small set of policies, runs in CPU minutes, and we release it as a named methodological contribution. Third, the

Table 1: Oracle-null results (unified protocol: within-question permutation of 4 policy correctness labels; gap = per-question oracle – best fixed policy; 10k iterations, seed 2024). High p = observed gap not in the upper tail.

Benchmark	System	Random floor	Oracle	p (one-sided)
RVS-Ego (314q)	42.7	42.8	51.6	0.989
StreamingBench (180q)	73.9	69.6	76.7	0.877
OVO-Bench (180q)	39.4	38.5	45.6	0.975

honest residual: the system sits *at* the random-selector floor on RVS (42.7 vs 42.8), above it on StreamingBench (+4.3) and OVO (+0.9) — a benchmark-dependent planner effect that Section 5 explains.

5 The Diagnosis: Coverage Prerequisite, Temporal Binding

5.1 A monotone correlation across three benchmarks

The planner’s gain over random policy selection ($-0.1 / +0.9 / +4.3$ on RVS / OVO / StreamingBench) tracks mean window coverage (0.21 / 0.59 / 1.00) almost perfectly — monotone in the thesis direction on both the full-question and all-wrong populations (correlational, three points; reported as a correlation, not a law).

5.2 The ceiling decomposition

Partitioning every question by whether any grain policy answers it correctly: RVS splits 110 all-right / 54 mixed / 152 all-wrong (48%); StreamingBench 113 / 25 / 42 (23%). Decomposing the all-wrong set: **zero retrieval failures** (no case where the evidence was present and unretrieved), 14 store-coverage failures, and a dominant model-capability remainder — with window-coverage evidence (mean 0.21 vs 0.071 chance; 83/138 temporal-before/after) localizing the capability failure to *long-window temporal grounding*: the model cannot integrate evidence across a wide temporal span even when the store holds it. StreamingBench’s all-wrong set is capability-failure by construction (its windows are short and end at the stream frontier, where coverage is 1.0).

5.3 The causal test: refuted on evidence

We attempted to force the diagnosis: four allocation policies (top-k, uniform-spread, hybrid, stratified) over one store and one 64-slot budget, spread computed strictly over streamed history (never the gold window — asserted in code), on the 83 temporal questions, with a pre-registered decision mapping. The deciding two-arm test was underpowered ($p=0.180$ vs a pre-declared MDE of 16–21 points); we therefore ran the powered version: a pooled regression over all 332 (question, arm) observations with question fixed effects and cluster-robust errors. **The result is a refutation, not a null-of-power:** the coverage coefficient is flat and insignificant in

Table 2: In-harness standings (7B, RVS-Ego, 72B judge; accuracy paired-tested; storage at true 0.5 fps, fp16). MuKV/ReKV storage from their papers.

System	Acc	GB/h	Paired vs DECAF
ReKV-7B (full KV, offload) [3]	54.1	18.8	-1.7, $p=0.139$
DECAF multi-grain (writetime)	52.4	10.38	—
MuKV-7B (faithful config) [11]	50.9	0.91–1.23	+1.5, $p=0.203$
best single grain (patch)	45.6	—	$\ll$

Pareto figure placeholder — accuracy vs GB/h at true 0.5 fps: DECAF arms (tax sweep, int4 variants as measured), MuKV, ReKV-7B, single-grain policies. Regenerated from run artifacts at camera-ready; ships whatever the W5.3 outcome.

Figure 1: Accuracy vs storage at true 0.5 fps (placeholder; final data pending the W5 gate).

every specification (question-FE: -0.33 per unit coverage, CI $[-1.99, +1.33]$), the coverage–accuracy curve is non-monotone across six bins, and the one arm with a real accuracy gain (hybrid, +6 pt) had *lower* coverage than the baseline. Within the achieved coverage band (≈ 0.18 – 0.49 — the arms spread retrieval *content*, Jaccard 0.18 – 0.29 , but barely coverage, cross-arm correlation 0.94 – 0.99), **coverage is not the lever**. Combined with the oracle-null result, the closure is complete: on this system, these questions are answered wrong regardless of which evidence the allocator selects — the binding constraint is model capability at long temporal windows, and no allocation mechanism at this store budget reaches the coverage where that might change.

5.4 What the field is optimizing

Grain selection, bit-width allocation, and retrieval ordering — the literature’s active axes — are second-order terms twice over: the oracle headroom they chase is an artifact (Section 4), and the coverage lever that might have mattered is refuted (this section). The binding constraint is the model’s long-window temporal grounding, which current streaming benchmarks’ standard splits barely probe and current mechanisms do not address. That is where an accuracy claim could live, and it is out of scope for a measurement paper.

6 The Corrected Pareto Accounting

6.1 The frame-rate correction

An autopsy of our own sweep found the npy video path double-resampled to an effective 0.25 fps against 0.5-fps anchors. Corrected at true 0.5 fps (standing assertion now live): the multi-grain store’s accuracy rises $51.1 \rightarrow 52.4$ ($+1.3$, $p=0.093$ — the extra frames genuinely help), and **store bytes double to 10.38 GB/h** (fixed per-segment token budgets times twice the segments). The pre-fix storage ratio had quietly compared our half-rate store against competitors’ full-rate numbers; we retract it.

6.2 Where the frontier stands

After multiplicity correction over the full 21-comparison family: **no accuracy claim survives** — the honest statement is that DECAF matches CVPR’26 SOTA accuracy and is statistically indistinguishable from full-fidelity ReKV. The storage position is likewise honest: 8.4 – $11.4\times$ MuKV’s published storage in fp16; the int4-group-scale repair (answer-level verified: naive int4 flips 4/5 greedy answers; group scales pending

measurement) is the only route to parity, and we report its outcome separately rather than assuming $4\times$.

6.3 The pre-registered Pareto attempt

Whether a low-tax operating point (corrected ≈ 5.6 GB/h) plus verified int4 group scales reaches storage parity at matched accuracy is an engineering question with a pre-registered gate; its outcome strengthens or closes this section but does not change the paper. Whatever it yields, the claim ceiling is fixed in advance: *matches SOTA accuracy at comparable storage with a different architecture*. We do not write “beats” for any margin that fails Holm — in any document, at any point.

7 Discussion

Threats to validity. The 72B judge substitutes for the field’s GPT judges; the stability argument (three systems, consistent offset) bounds but does not eliminate this risk, and letter-match benchmarks bypass it entirely. The RVS oracle analysis rests on a 2-video subset (mitigated by the three-benchmark protocol). The coverage–gain correlation is three points. The causal coverage experiment is underpowered at $n=83$ by pre-declared MDE. Each caveat is load-bearing and stated where the number lives.

What we would tell a new entrant. Do not build a mechanism against an oracle bound until you have run the oracle-null protocol on it. Do not trust a repository’s defaults against its paper. Assert your effective frame rate. Label your statistical tails at emit time. Pre-register your primary comparison and your multiplicity family, and apply the correction to your own claims before a reviewer does.

Ethics and scope. All experiments use public benchmarks; no human subjects. The measurement stack (code, judged artifacts, statistical tooling) is released for reuse.

8 Conclusion

We set out to beat the strongest streaming video-LLM memory system and built the instrument that would prove it. The instrument is the result: faithful reproductions that exposed a config-fidelity trap in a CVPR’26 system’s own repository, a judge whose stability across three systems makes a literature comparable, a pre-registration discipline that killed six mechanisms — including our own flagship — and an oracle-null protocol that retires a class of motivating bound the field has been quoting uncritically. What survives is a

corrected frontier (match, not beat, at an honestly-measured storage cost), a monotone coverage–gain correlation, and a diagnosis that names the real problem: long-window temporal grounding. The next mechanism worth building is the one that attacks that.

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A Reproducibility Trail

Every number in this paper traces to a run directory containing `config.json` (written before execution), `env.json`, `code_state.json` (commits + patches), `fps_assertion.jsonl`, metrics, and the judged artifact. Index (server: Ada workspace; mirrored to the paper’s repository): Phase 0 Gate 0 MuKV reproduction (20260927_0000); Phase 1 Component A and its corrected rank-fetch confirmation (20260927_1200, 20260927_2200); Phase 2 Component B sweep, oracle chain, W3 hardening analysis (20260928_0315, analysis/w31,w32,w33,w41,w43,w54); cross-benchmark oracles (20260928_2105, 20260929_1215); W4 frame-rate correction and coverage-spread experiment (20260930_0935). Code: `decaf` repository, commits 57ae130–93014b3 plus W4/W5 fix commits. The append-only experiment log records every run including failures and superseded analyses.

B Additional Statistical Detail

Gate 0 config fidelity. MuKV argparse-default run: 46.0 (band [47.5, 53.5], FAIL); paper-config run: 50.9 (PASS, expected ≈ 50.5 under the calibrated offset); minimal-pruning sanity: 51.2 (code-path validation). The paper configuration (segment-dominant keep 0.1/0.1/0.8, per-grain topk 20/32/12, α 0.5/0.7/0.8) is now encoded in a versioned script.

W5.4 pooled regression (full). $n=332$ (4 arms $\times$ 83 temporal questions); LPM with cluster-robust standard errors (cluster = question). Specifications: question FE: coefficient -0.33 per unit coverage, CI $[-1.99, +1.33]$, $p=0.69$; arm FE + window length: $+0.56$, CI $[-0.54, +1.66]$, $p=0.32$; question FE + arm FE: $+0.40$, CI $[-1.33, +2.13]$, $p=0.65$. Six-bin coverage–accuracy curve non-monotone; the hybrid arm’s accuracy gain coincided with *lower* coverage than baseline. Within-question coverage variation (sd 0.019) is small relative to between-question (0.070); the refutation is conditional on the achieved band and store budget, which the text states.

Answer-level int4 (TB6). Naive per-channel int4 quantization of the store flips 4 of 5 greedy answers at equal prompts; logit-level fidelity (the basis of prior “4-bit is near-lossless” claims) does not predict answer-level fidelity. Group-scale int4 is measured at answer level wherever it is reported.

Multiplicity families. Family A ($m=21$): all pairs among the seven full-run RVS arms. Family B ($m=15$): the pre-registered comparison subset. Union $m=36$. All corrected claims in the text are stated against the family; Holm primary, BH reported.